



Incident handler's journal

Date: June-22-2026	Entry: 1
Description	A healthcare clinic suffered a ransomware attack after an employee opened a phishing email containing malicious attachment. The malware allowed attackers to access the network and access critical patient files, disrupting business operations.
Tool(s) used	None
The 5 W's	Capture the 5 W's of an incident. ● Who caused the incident? Organized group of unethical hackers ● What happened? Attackers sent a phishing email with malicious attachment. After an employee downloaded the attachment, malware was installed, allowing attackers to access the network and deploy ransomware that encrypted important files and medical records. ● When did the incident occur? On a Tuesday morning, approximately at 9:00 am ● Where did the incident happen? At US health care clinic ● Why did the incident happen? The employee was targeted with a phishing email. Malicious attachment was downloaded, giving attackers access to a network

	where they deploy ransomware to encrypt files and demand payment for decryption keys.
Additional notes	• The organization's reputation may have been damaged because patients could lose trust in its ability to protect sensitive medical information. • Employees may not have had sufficient cybersecurity awareness training to recognize phishing emails and avoid downloading malicious attachments. • Patient records became inaccessible due to ransomware encryption, which could have delayed or disrupted patient treatment and healthcare services. • The organization should improve employee security training, email filtering, and data backup procedures to reduce the risk of future attacks

Date: June-25-2026	Entry: 2
Description	This journal entry documents the investigation of a suspicious file downloaded from a phishing email attachment. Using the file's SHA256 hash, the file was analyzed in VirusTotal, confirmed as malicious, and several indicators of compromise (IoCs) were identified to support the incident investigation.
Tool(s) used	VirusTotal

	Intrusion Detection System (IDS)
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? An unknown threat actor who sent a malicious email attachment. • What happened? An employee opened a malicious spreadsheet file, which executed malware on the computer. • When did the incident occur? Between 1:11 p.m. and 1:20 p.m., when the email was received, opened, and detected. • Where did the incident happen? On an employee's workstation within the company's network. • Why did the incident happen? The employee opened a malicious attachment received through a phishing email.
Additional notes	• Password-protected attachment was used to bypass security controls. • Malware executed after the employee opened the file. • Unauthorized executable files were created on the system. • Check if other devices or accounts were affected. • Investigate for additional indicators of compromise (IoCs).

Date:	Entry:
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June-25-2026	3
Description	This journal entry documents the investigation and resolution of a phishing alert involving a malicious email attachment. The file hash was analyzed and confirmed to be malicious, and the incident was handled according to the organization's phishing response playbook. The findings were recorded and the alert ticket was updated with the investigation results.
Tool(s) used	Phishing Response Playbook Alert Ticketing System
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? An unknown attacker sent a phishing email. • What happened? An employee opened a malicious email attachment, causing malware to execute. • When did the incident occur? The incident occurred between 1:11 p.m. and 1:20 p.m. • Where did the incident happen? On an employee's workstation within the company network. • Why did the incident happen? The employee opened a malicious attachment delivered through a phishing email.
Additional notes	• The email attachment was confirmed to be malicious through VirusTotal analysis. • The incident was handled according to the organization's phishing response playbook.

	• The alert ticket was updated with investigation findings and recommended actions. • User awareness training may help prevent similar phishing incidents in the future. • Continue monitoring for any additional suspicious activity related to this incident.
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Date: June-25-2026.	Entry: 4
Description	This journal entry documents the investigation of a data breach in which an attacker exploited a vulnerability in the organization's e-commerce web application to gain unauthorized access to customer PII and financial information. The security team identified the root cause, assessed the impact on approximately 50,000 customer records, and implemented remediation measures to prevent future incidents.
Tool(s) used	Log Analysis Tools Web Server Logs
The 5 W's	• Who caused the incident? An unknown attacker who exploited a web application vulnerability. • What happened?

	The attacker gained unauthorized access to customer PII and financial information and exfiltrated the data • When did the incident occur? The breach was discovered on December 28, 2022, after a follow-up extortion email was received. The initial email was received on December 22, 2022. • Where did the incident happen? In the organization's e-commerce web application • Why did the incident happen? A forced browsing vulnerability allowed the attacker to modify order numbers in URLs and access customer transaction data without authorization.
Additional notes	• Approximately 50,000 customer records were affected by the breach. • The attacker exploited a forced browsing vulnerability in the web application. • Web application and server logs showed a high volume of sequential requests to customer order pages. • Regular vulnerability scanning, penetration testing, and stronger access controls are recommended to prevent similar incidents.

Date: June 25-2026	Entry: 5
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Description	Used tcpdump to identify network interfaces, capture live HTTP traffic, save captured packets to a .pcap file, and inspect packet headers and payloads. The captured data was filtered and reviewed to analyze network communications.
Tool(s) used	tcpdump
NIST phase	Detection and Analysis
Reason	tcpdump was used to capture and inspect network packets, supporting the detection and analysis of network activity during a security investigation.

Date: July 1- 2026	Entry: 6
Description	Used Suricata to test a custom detection rule (sid:12345) against sample.pcap via <code>sudo suricata -r sample.pcap -S custom.rules -k none</code> , then reviewed the results in both fast.log and eve.json, using jq to parse and extract key fields (timestamp, flow_id, alert signature, protocol, destination IP) from the JSON output for easier analysis.
Tool(s) used	Suricata (Network Intrusion Detection System)
NIST phase	Preparation and Detection and Analysis — configuring and testing the rule falls under Preparation, while running it against traffic and reviewing the resulting alerts falls under Detection and Analysis.

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Reflections

1. Were there any specific activities that were challenging for you? Why or why not?

I did not find any activity particularly difficult or frustrating. I carefully read the course documentation and watched all the instructional videos before completing each lab. Whenever I encountered a challenge, I referred back to the course materials to better understand the concept and then successfully completed the activity.

2. Has your understanding of incident detection and response changed since taking this course?

Yes, this course has significantly improved my understanding of incident detection and response. Learning new cybersecurity concepts and applying them through hands-on labs helped me understand how security analysts detect, investigate, and respond to potential security incidents in real-world scenarios.

3. Was there a specific tool or concept that you enjoyed the most? Why?

I especially enjoyed working with Chronicle and Splunk. I liked searching logs using different filters and queries, then analyzing the results to identify relevant events and patterns. These activities gave me practical experience in log analysis and helped me better understand how security investigations are performed.